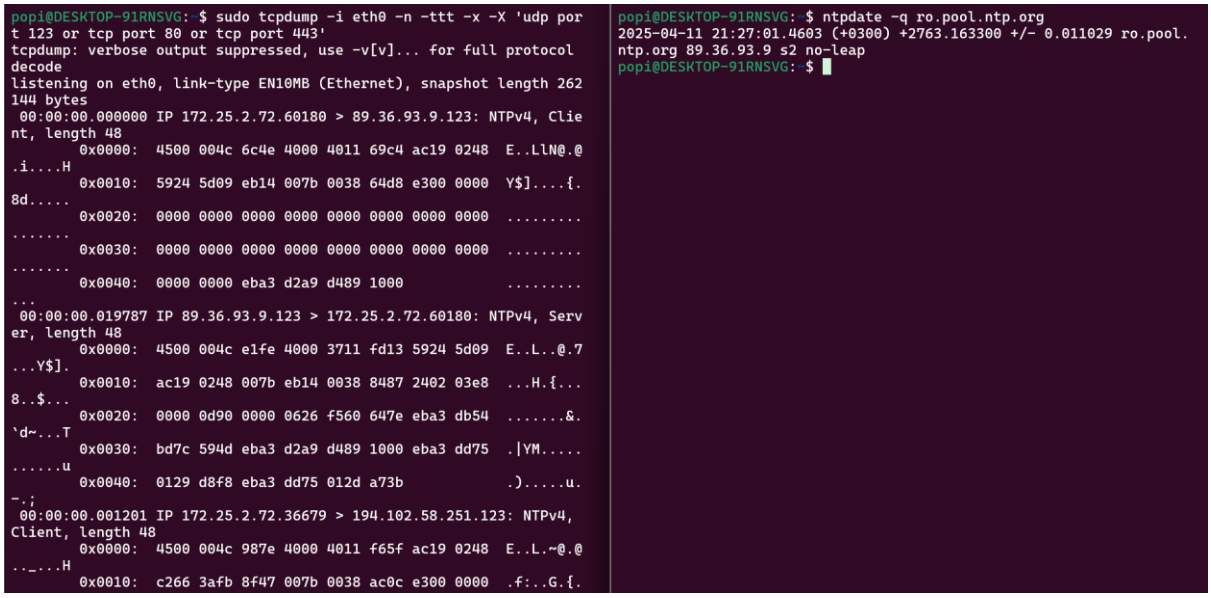
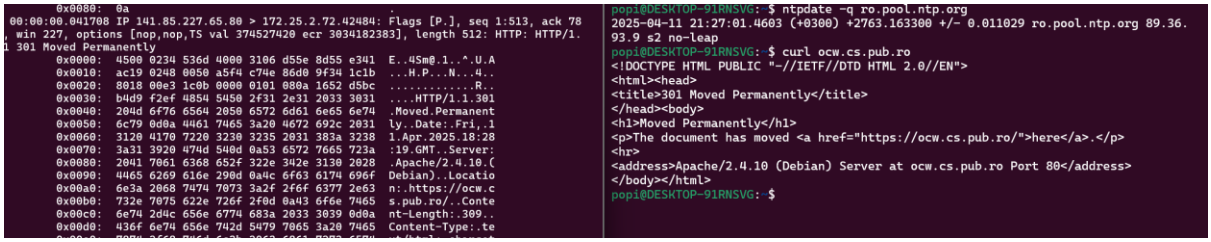
POPESCU MARIA-TEODORA
345C1
FACULTATEA DE AUTOMATICA SI CALCULATOARE – SPECIALIZAREA CALCULATOARE
UNIVERSITATEA NATIONALA DE STIINTA SI TEHNOLOGIE POLITEHNICA DIN BUCURESTI

Laborator 6 – EP

1.

A.





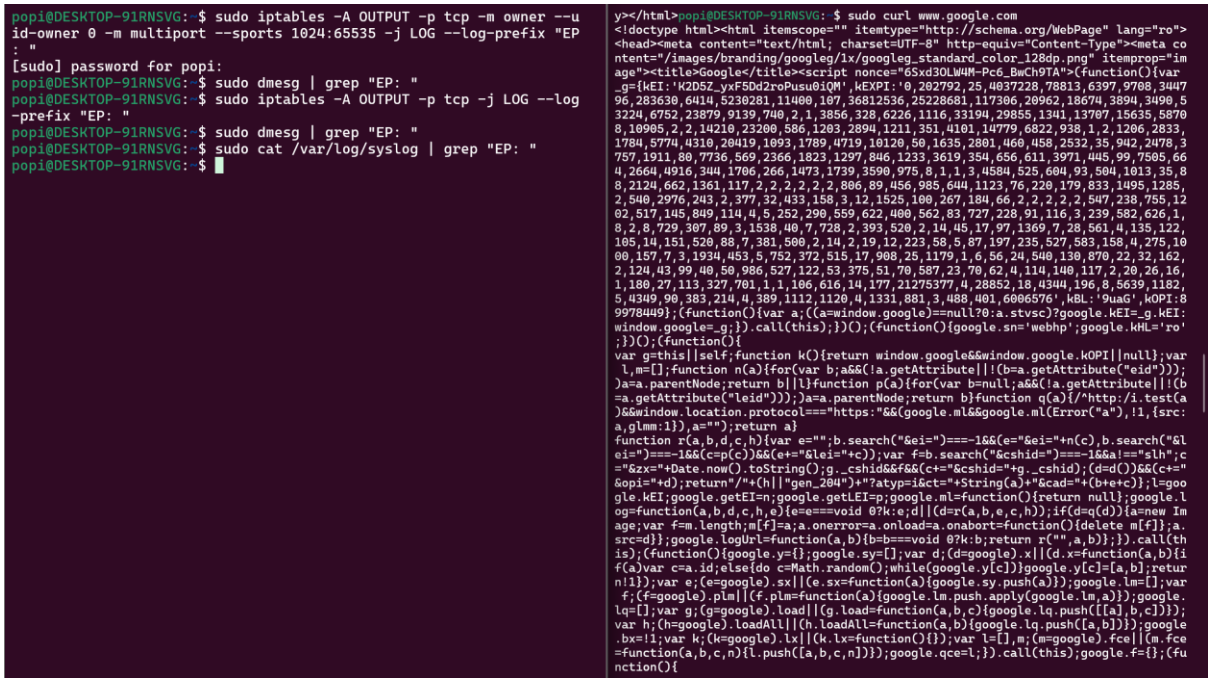
B.

1.

sudo iptables -A OUTPUT -p tcp -m owner --uid-owner 0 -m multiport --sports 1024:65535 -j LOG --log-prefix "EP: "

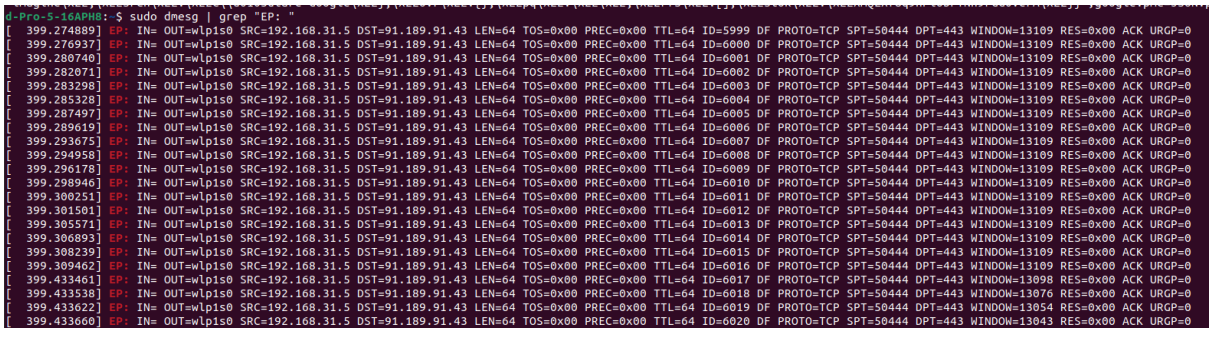
În mod normal, această comandă ar adăuga o regulă iptables care identifică pachetele TCP originare din porturile efemere (1024-65535) create de utilizatorul root și le adaugă la log-urile kernel-ului cu prefixul 'EP: '. După rularea sudo curl www.google.com, ar trebui să apară mesaje în dmesg care conțin acest prefix, indicând că pachetele TCP generate de comanda curl (rulată ca root) au fost identificate și logate. Dar la mine nu inteleg de ce nu merge..... Ma gandesc ca există niste limitări în WSL 2 legate de funcționalitățile iptables și log-urile kernel-

ului, ceea ce poate face ca unele operații standard Linux să nu funcționeze exact la fel ca pe un sistem Linux nativ.



// incerc imediat si pe linux-linux, nu pe wsl – A MERS =D





2.

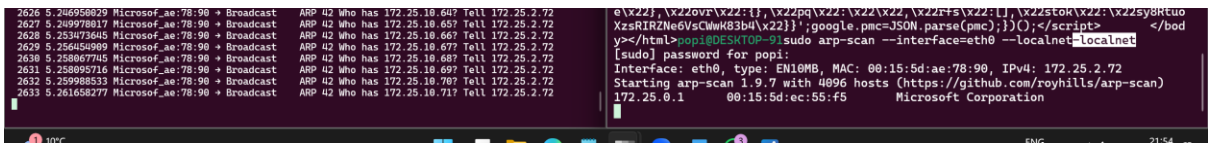
```
popi@DESKTOP-91RNSVG:~$ UDP_53_FILTER=$(sudo nfbpf_compile 'udp dst port 53')
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -m bpf --bytecode "$UDP_5
3_FILTER" -j TTL --ttl-set 1
iptables v1.8.7 (nf_tables): Couldn't load match `bpf':No such file or directory

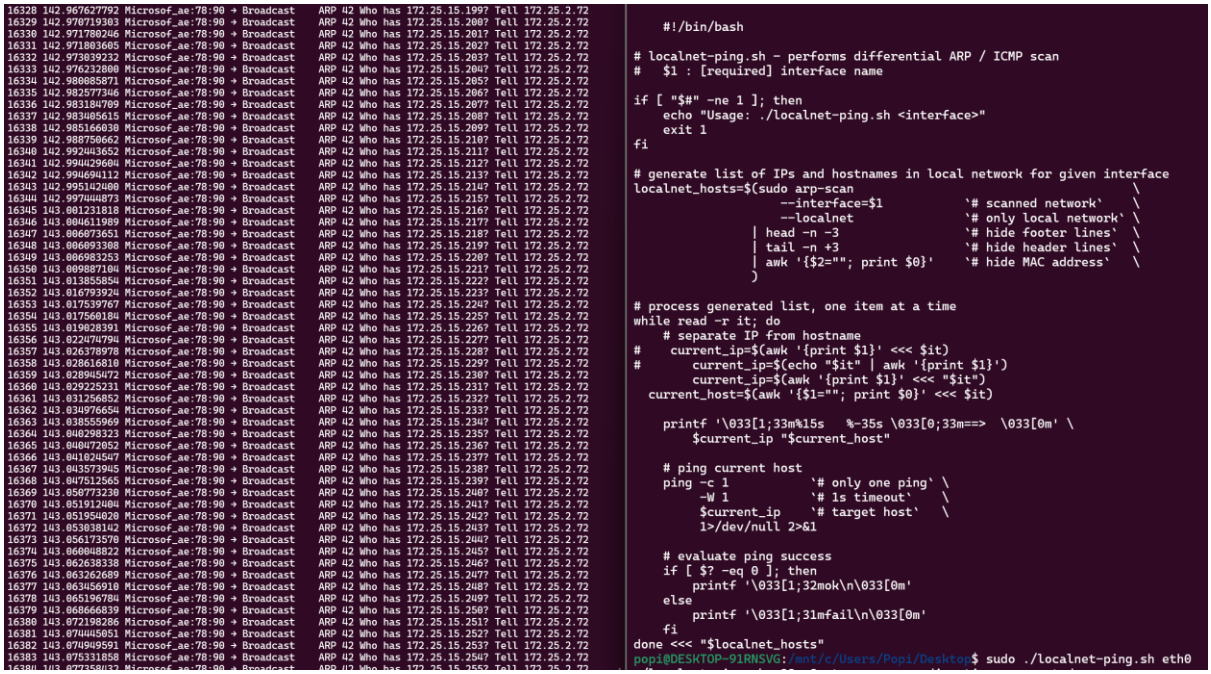
Try `iptables -h' or 'iptables --help' for more information.
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 1
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; no servers could be reached
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 2
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; no servers could be reached
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 4
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; no servers could be reached
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 8
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; no servers could be reached
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 16
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
141.85.150.247
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 14
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
141.85.150.247
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 12
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
^[[A^[[B;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; communications error to 8.8.8.8#53: timed out
;; no servers could be reached
popi@DESKTOP-91RNSVG:~$ sudo iptables -t mangle -A OUTPUT -p udp --dport 53 -j TTL --ttl-set 13
popi@DESKTOP-91RNSVG:~$ dig +short fep.grid.pub.ro @8.8.8.8
141.85.150.247
popi@DESKTOP-91RNSVG:~$
```

3. Iptables nu poate captura pachete care sunt procesate la nivel de hardware sau de driver, înainte ca acestea să ajungă la stiva de rețea a kernel-ului. De exemplu, pachetele procesate de hardware offloading sau pachete care sunt gestionate la nivelul Layer 2 (cum ar fi pachetele ARP) nu pot fi capturate de iptables, care operează la nivel IP (Layer 3).

2.

A.





```
16328 142.967627792 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.199? Tell 172.25.2.72
16329 142.970719303 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.200? Tell 172.25.2.72
16330 142.971780246 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.201? Tell 172.25.2.72
16331 142.971803605 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.202? Tell 172.25.2.72
16332 142.973039232 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.203? Tell 172.25.2.72
16333 142.976232800 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.204? Tell 172.25.2.72
16334 142.980085871 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.205? Tell 172.25.2.72
16335 142.982577346 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.206? Tell 172.25.2.72
16336 142.983184709 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.207? Tell 172.25.2.72
16337 142.983405615 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.208? Tell 172.25.2.72
16338 142.985166030 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.209? Tell 172.25.2.72
16339 142.988750662 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.210? Tell 172.25.2.72
16340 142.992443652 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.211? Tell 172.25.2.72
16341 142.994429604 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.212? Tell 172.25.2.72
16342 142.994694112 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.213? Tell 172.25.2.72
16343 142.995142408 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.214? Tell 172.25.2.72
16344 142.997444873 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.215? Tell 172.25.2.72
16345 143.001231818 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.216? Tell 172.25.2.72
16346 143.004611989 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.217? Tell 172.25.2.72
16347 143.006073651 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.218? Tell 172.25.2.72
16348 143.006093308 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.219? Tell 172.25.2.72
16349 143.006983253 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.220? Tell 172.25.2.72
16350 143.009887104 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.221? Tell 172.25.2.72
16351 143.013855854 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.222? Tell 172.25.2.72
16352 143.016793924 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.223? Tell 172.25.2.72
16353 143.017539767 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.224? Tell 172.25.2.72
16354 143.017560184 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.225? Tell 172.25.2.72
16355 143.019028391 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.226? Tell 172.25.2.72
16356 143.022474794 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.227? Tell 172.25.2.72
16357 143.026378978 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.228? Tell 172.25.2.72
16358 143.028616810 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.229? Tell 172.25.2.72
16359 143.028945472 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.230? Tell 172.25.2.72
16360 143.029225231 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.231? Tell 172.25.2.72
16361 143.031256852 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.232? Tell 172.25.2.72
16362 143.034976654 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.233? Tell 172.25.2.72
16363 143.038555969 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.234? Tell 172.25.2.72
16364 143.040298323 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.235? Tell 172.25.2.72
16365 143.040472052 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.236? Tell 172.25.2.72
16366 143.041024547 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.237? Tell 172.25.2.72
16367 143.043573945 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.238? Tell 172.25.2.72
16368 143.047512565 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.239? Tell 172.25.2.72
16369 143.050773230 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.240? Tell 172.25.2.72
16370 143.051912404 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.241? Tell 172.25.2.72
16371 143.051954020 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.242? Tell 172.25.2.72
16372 143.053038142 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.243? Tell 172.25.2.72
16373 143.056173570 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.244? Tell 172.25.2.72
16374 143.060048822 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.245? Tell 172.25.2.72
16375 143.062638338 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.246? Tell 172.25.2.72
16376 143.063262689 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.247? Tell 172.25.2.72
16377 143.063456910 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.248? Tell 172.25.2.72
16378 143.065196784 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.249? Tell 172.25.2.72
16379 143.068666839 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.250? Tell 172.25.2.72
16380 143.072198286 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.251? Tell 172.25.2.72
16381 143.074445051 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.252? Tell 172.25.2.72
16382 143.074949591 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.253? Tell 172.25.2.72
16383 143.075331858 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.254? Tell 172.25.2.72
```

```
    #!/bin/bash

# localnet-ping.sh - performs differential ARP / ICMP scan
#   $1 : [required] interface name

if [ "$#" -ne 1 ]; then
    echo "Usage: ./localnet-ping.sh <interface>"
    exit 1
fi

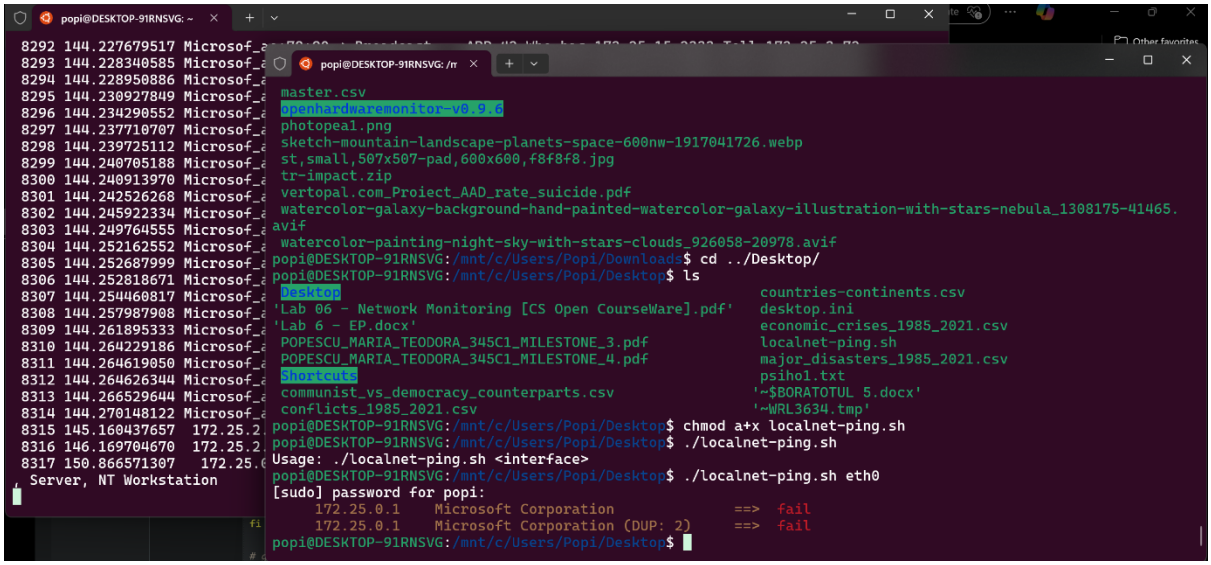
# generate list of IPs and hostnames in local network for given interface
localnet_hosts=$(sudo arp-scan                          \
                    --interface=$1              `# scanned network`     \
                    --localnet                  `# only local network` \
                | head -n -3                    `# hide footer lines`  \
                | tail -n +3                    `# hide header lines`  \
                | awk '{$2=""; print $0}'       `# hide MAC address`   \
                )

# process generated list, one item at a time
while read -r ip; do
    # separate IP from hostname
#    current_ip=$(awk '{print $1}' <<< $ip)
#       current_ip=$(echo "$ip" | awk '{print $1}')
        current_ip=$(awk '{print $1}' <<< "$ip")
  current_host=$(awk '{$1=""; print $0}' <<< $ip)

    printf '\033[1;33m%15s   %-35s \033[0;33m==>  \033[0m' \
        $current_ip "$current_host"

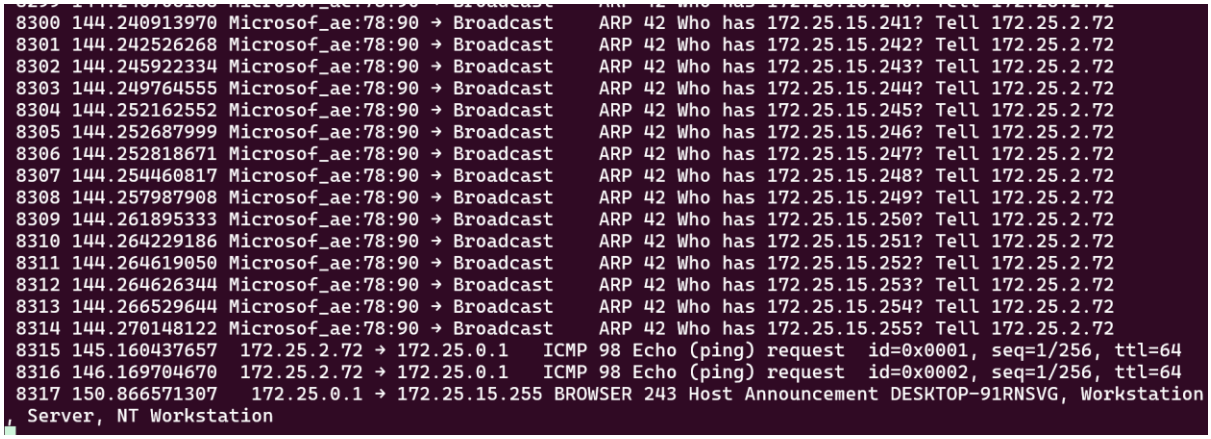
    # ping current host
    ping -c 1           `# only one ping` \
         -W 1           `# 1s timeout`    \
         $current_ip    `# target host`   \
         1>/dev/null 2>&1

    # evaluate ping success
    if [ $? -eq 0 ]; then
        printf '\033[1;32mok\n\033[0m'
    else
        printf '\033[1;31mfail\n\033[0m'
    fi
done <<< "$localnet_hosts"
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo ./localnet-ping.sh eth0
```



```
popi@DESKTOP-91RNSVG: ~
 8292 144.227679517 Microsof_a
 8293 144.228340585 Microsof_a
 8294 144.228950886 Microsof_a
 8295 144.230927849 Microsof_a
 8296 144.234290552 Microsof_a
 8297 144.237710707 Microsof_a
 8298 144.239725112 Microsof_a
 8299 144.240705188 Microsof_a
 8300 144.240913970 Microsof_a
 8301 144.242526268 Microsof_a
 8302 144.245922334 Microsof_a
 8303 144.249764555 Microsof_a
 8304 144.252162552 Microsof_a
 8305 144.252687999 Microsof_a
 8306 144.252818671 Microsof_a
 8307 144.254460817 Microsof_a
 8308 144.257987908 Microsof_a
 8309 144.261895333 Microsof_a
 8310 144.264229186 Microsof_a
 8311 144.264619050 Microsof_a
 8312 144.264626344 Microsof_a
 8313 144.266529644 Microsof_a
 8314 144.270148122 Microsof_a
 8315 145.160437657  172.25.2
 8316 146.169704670  172.25.2
 8317 150.866571307   172.25.0
, Server, NT Workstation
```

```
popi@DESKTOP-91RNSVG: /mnt
 master.csv
 openhardwaremonitor-v0.9.6
 photopea1.png
 sketch-mountain-landscape-planets-space-600nw-1917041726.webp
 st,small,507x507-pad,600x600,f8f8f8.jpg
 tr-impact.zip
 vertopal.com_Proiect_AAD_rate_suicide.pdf
 watercolor-galaxy-background-hand-painted-watercolor-galaxy-illustration-with-stars-nebula_1308175-41465.
avif
 watercolor-painting-night-sky-with-stars-clouds_926058-20978.avif
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Downloads$ cd ../Desktop/
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ ls
 Desktop                                        countries-continents.csv
'Lab 06 - Network Monitoring [CS Open CourseWare].pdf'   desktop.ini
'Lab 6 - EP.docx'                               economic_crises_1985_2021.csv
 POPESCU_MARIA_TEODORA_345C1_MILESTONE_3.pdf    localnet-ping.sh
 POPESCU_MARIA_TEODORA_345C1_MILESTONE_4.pdf    major_disasters_1985_2021.csv
 Shortcuts                                      psiho1.txt
 communist_vs_democracy_counterparts.csv       '~$BORATOTUL 5.docx'
 conflicts_1985_2021.csv                       '~WRL3634.tmp'
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ chmod a+x localnet-ping.sh
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ ./localnet-ping.sh
Usage: ./localnet-ping.sh <interface>
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ ./localnet-ping.sh eth0
[sudo] password for popi:
     172.25.0.1    Microsoft Corporation             ==>  fail
     172.25.0.1    Microsoft Corporation (DUP: 2)    ==>  fail
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
```



```
 8300 144.240913970 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.241? Tell 172.25.2.72
 8301 144.242526268 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.242? Tell 172.25.2.72
 8302 144.245922334 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.243? Tell 172.25.2.72
 8303 144.249764555 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.244? Tell 172.25.2.72
 8304 144.252162552 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.245? Tell 172.25.2.72
 8305 144.252687999 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.246? Tell 172.25.2.72
 8306 144.252818671 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.247? Tell 172.25.2.72
 8307 144.254460817 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.248? Tell 172.25.2.72
 8308 144.257987908 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.249? Tell 172.25.2.72
 8309 144.261895333 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.250? Tell 172.25.2.72
 8310 144.264229186 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.251? Tell 172.25.2.72
 8311 144.264619050 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.252? Tell 172.25.2.72
 8312 144.264626344 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.253? Tell 172.25.2.72
 8313 144.266529644 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.254? Tell 172.25.2.72
 8314 144.270148122 Microsof_ae:78:90 → Broadcast    ARP 42 Who has 172.25.15.255? Tell 172.25.2.72
 8315 145.160437657  172.25.2.72 → 172.25.0.1    ICMP 98 Echo (ping) request  id=0x0001, seq=1/256, ttl=64
 8316 146.169704670  172.25.2.72 → 172.25.0.1    ICMP 98 Echo (ping) request  id=0x0002, seq=1/256, ttl=64
 8317 150.866571307   172.25.0.1 → 172.25.15.255 BROWSER 243 Host Announcement DESKTOP-91RNSVG, Workstation
, Server, NT Workstation
```

B.

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo arp-scan --interface=eth0 --localnet
Interface: eth0, type: EN10MB, MAC: 00:15:5d:ae:78:90, IPv4: 172.25.2.72
Starting arp-scan 1.9.7 with 4096 hosts (https://github.com/royhills/arp-scan)
172.25.0.1      00:15:5d:ec:55:f5       Microsoft Corporation

1 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.9.7: 4096 hosts scanned in 17.474 seconds (234.41 hosts/sec). 1 responded
```

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo nmap
    -sn     `# disable port scan`      \
    -Pn     `# disable host discovery` \
    -tr     `# perform traceroute`     \
    8.8.8.8
Starting Nmap 7.80 ( https://nmap.org ) at 2025-04-11 00:35 EEST
Nmap scan report for 8.8.8.8
Host is up (0.045s latency).

TRACEROUTE (using proto 1/icmp)
HOP RTT      ADDRESS
1   0.57 ms  DESKTOP-91RNSVG (172.25.0.1)
2   32.54 ms GatewayA1Etaj9 (172.21.109.1)
3   32.55 ms 94.177.34.1
4   33.15 ms 105.4.18.89.in-addr.arpa (89.18.4.105)
5   32.56 ms ae3.10-edge1.buh.primetelecom.ro (89.44.235.101)
6   32.56 ms 212.133.89.61
7   43.80 ms ae1.3101.ear1.sof5.neo.colt.net (171.75.9.241)
8   44.88 ms 173.194.121.16
9   44.87 ms 142.251.244.109
10  53.29 ms 172.253.73.21
11  44.98 ms 8.8.8.8

Nmap done: 1 IP address (1 host up) scanned in 26.18 seconds
```

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tshark -i eth0 -w nmap_capture.pcap -f "host 8.8
.8.8" & TSHARK_PID=$!
[2] 9577
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ Running as user "root" and group "root". This could b
e dangerous.
Capturing on 'eth0'
 ** (tshark:9579) 21:17:27.854325 [Main MESSAGE] -- Capture started.
 ** (tshark:9579) 21:17:27.854468 [Main MESSAGE] -- File: "nmap_capture.pcap"
30 16395 449.857232170 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 Who has 172.25.2.72? Tell 172.25.
0.1
16396 449.857369368 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 172.25.2.72 is at 00:15:5d:ae:78:90
16397 450.071345372 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 Who has 172.25.0.1? Tell 172.25.2.72
16398 450.071653364 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 172.25.0.1 is at 00:15:5d:ec:55:f5
^C
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tshark -i eth0 -w traceroute_capture.pcap -f "ho
st 8.8.8.8" & TSHARK_PID=$!
[3] 9629
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ Running as user "root" and group "root". This could b
e dangerous.
Capturing on 'eth0'
 ** (tshark:9631) 21:18:13.504316 [Main MESSAGE] -- Capture started.
 ** (tshark:9631) 21:18:13.504414 [Main MESSAGE] -- File: "traceroute_capture.pcap"
32 16399 488.871804126 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 Who has 172.25.2.72? Tell 172.25.
0.1
16400 488.871956488 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 172.25.2.72 is at 00:15:5d:ae:78:90
62 16401 489.717021827 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 Who has 172.25.0.1? Tell 172.25.2
.72
16402 489.717248617 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 172.25.0.1 is at 00:15:5d:ec:55:f5
58
```

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo nmap -sn -Pn -tr 8.8.8.8
Starting Nmap 7.80 ( https://nmap.org ) at 2025-04-11 21:17 EEST
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.029s latency).

TRACEROUTE (using proto 1/icmp)
HOP RTT      ADDRESS
1   0.58 ms  DESKTOP-91RNSVG (172.25.0.1)
2   27.05 ms 192.168.85.152
3   27.07 ms XiaoQiang.lan (192.168.31.1)
4   27.13 ms 192.168.0.1
5   27.12 ms 10.0.2.58
6   27.22 ms 10.243.116.97
7   27.21 ms 10.220.202.70
8   43.49 ms 10.220.131.95
9   47.77 ms 10.221.100.14
10  47.99 ms 209.85.168.182
11  37.25 ms 209.85.250.89
12  29.51 ms 142.251.65.225
13  29.41 ms dns.google (8.8.8.8)

Nmap done: 1 IP address (1 host up) scanned in 1.48 seconds
```

Am incercat si o analiza a capturilor

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  DESKTOP-91RNSVG (172.25.0.1)  0.902 ms  0.651 ms  0.552 ms
 2  192.168.85.152 (192.168.85.152)  28.741 ms  28.670 ms  28.591 ms
 3  XiaoQiang.lan (192.168.31.1)  28.511 ms  28.256 ms  28.142 ms
 4  192.168.0.1 (192.168.0.1)  28.193 ms  28.056 ms  27.969 ms
 5  10.0.2.58 (10.0.2.58)  27.907 ms  29.777 ms  27.614 ms
 6  10.243.116.97 (10.243.116.97)  32.269 ms  13.053 ms  10.512 ms
 7  10.220.202.70 (10.220.202.70)  16.105 ms  39.460 ms  39.564 ms
 8  10.220.131.97 (10.220.131.97)  46.382 ms 10.220.131.95 (10.220.131.95)  46.29
5 ms 10.220.148.206 (10.220.148.206)  45.827 ms
 9  10.221.100.14 (10.221.100.14)  99.791 ms  45.602 ms  99.143 ms
10  209.85.168.182 (209.85.168.182)  98.855 ms 72.14.216.212 (72.14.216.212)  44.
487 ms  43.814 ms
11  * * *
12  142.251.65.225 (142.251.65.225)  81.653 ms dns.google (8.8.8.8)  81.514 ms  4
0.751 ms
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tshark -r nmap_capture.pcap
-Y "ip.ttl < 64" -T fields -e ip.ttl -e ip.src -e ip.dst
Running as user "root" and group "root". This could be dangerous.
10      172.25.2.72     8.8.8.8
9       172.25.2.72     8.8.8.8
8       172.25.2.72     8.8.8.8
7       172.25.2.72     8.8.8.8
6       172.25.2.72     8.8.8.8
5       172.25.2.72     8.8.8.8
4       172.25.2.72     8.8.8.8
3       172.25.2.72     8.8.8.8
2       172.25.2.72     8.8.8.8
1       172.25.2.72     8.8.8.8
11      172.25.2.72     8.8.8.8
12      172.25.2.72     8.8.8.8
13      172.25.2.72     8.8.8.8
14      172.25.2.72     8.8.8.8
15      172.25.2.72     8.8.8.8
16      172.25.2.72     8.8.8.8
17      172.25.2.72     8.8.8.8
18      172.25.2.72     8.8.8.8
19      172.25.2.72     8.8.8.8
20      172.25.2.72     8.8.8.8
21      172.25.2.72     8.8.8.8
55      8.8.8.8 172.25.2.72
55      8.8.8.8 172.25.2.72
55      8.8.8.8 172.25.2.72
55      8.8.8.8 172.25.2.72
55      8.8.8.8 172.25.2.72
```

1

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tshark -r traceroute_capture.pcap -Y "ip.ttl < 64" -T fie
lds -e ip.ttl -e ip.src -e ip.dst
Running as user "root" and group "root". This could be dangerous.
1       172.25.2.72     8.8.8.8
1       172.25.2.72     8.8.8.8
1       172.25.2.72     8.8.8.8
2       172.25.2.72     8.8.8.8
2       172.25.2.72     8.8.8.8
2       172.25.2.72     8.8.8.8
3       172.25.2.72     8.8.8.8
3       172.25.2.72     8.8.8.8
3       172.25.2.72     8.8.8.8
4       172.25.2.72     8.8.8.8
4       172.25.2.72     8.8.8.8
4       172.25.2.72     8.8.8.8
5       172.25.2.72     8.8.8.8
5       172.25.2.72     8.8.8.8
5       172.25.2.72     8.8.8.8
6       172.25.2.72     8.8.8.8
6       172.25.2.72     8.8.8.8
6       172.25.2.72     8.8.8.8
7       172.25.2.72     8.8.8.8
7       172.25.2.72     8.8.8.8
7       172.25.2.72     8.8.8.8
8       172.25.2.72     8.8.8.8
8       172.25.2.72     8.8.8.8
8       172.25.2.72     8.8.8.8
9       172.25.2.72     8.8.8.8
9       172.25.2.72     8.8.8.8
9       172.25.2.72     8.8.8.8
10      172.25.2.72     8.8.8.8
10      172.25.2.72     8.8.8.8
10      172.25.2.72     8.8.8.8
11      172.25.2.72     8.8.8.8
11      172.25.2.72     8.8.8.8
11      172.25.2.72     8.8.8.8
12      172.25.2.72     8.8.8.8
12      172.25.2.72     8.8.8.8
55,1    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
12      172.25.2.72     8.8.8.8
13      172.25.2.72     8.8.8.8
13      172.25.2.72     8.8.8.8
13      172.25.2.72     8.8.8.8
14      172.25.2.72     8.8.8.8
14      172.25.2.72     8.8.8.8
14      172.25.2.72     8.8.8.8
15      172.25.2.72     8.8.8.8
15      172.25.2.72     8.8.8.8
16      172.25.2.72     8.8.8.8
55,1    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,2    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,3    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,1    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,2    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,4    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,4    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,4    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,3    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,2    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
55,2    8.8.8.8,172.25.2.72     172.25.2.72,8.8.8.8
```

Optional:

sudo tshark -i eth0 -w tcp_options.pcap -f "tcp[tcpflags] & (tcp-syn) != 0" & TSHARK_PID=$!

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ Running as user "root" and group "root". This could b
e dangerous.
Capturing on 'eth0'
 ** (tshark:9681) 21:23:48.994695 [Main MESSAGE] -- Capture started.
 ** (tshark:9681) 21:23:48.994822 [Main MESSAGE] -- File: "tcp_options.pcap"
2 16403 829.566869081 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 Who has 172.25.2.72? Tell 172.25.0
.1
16404 829.567092495 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 172.25.2.72 is at 00:15:5d:ae:78:90
```

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ curl google.com
<HTML><HEAD><meta http-equiv="content-type" content="text/html;charset=utf-8">
<TITLE>301 Moved</TITLE></HEAD><BODY>
<H1>301 Moved</H1>
The document has moved
<A HREF="http://www.google.com/">here</A>.
</BODY></HTML>
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tshark -r tcp_options.pcap -Y "tcp.flags.syn==1 and tcp.f
lags.ack==0" -O tcp | grep -A 20 "Options"
Running as user "root" and group "root". This could be dangerous.
    Options: (20 bytes), Maximum segment size, SACK permitted, Timestamps, No-Operation (NOP), Window scale
        TCP Option - Maximum segment size: 1460 bytes
            Kind: Maximum Segment Size (2)
            Length: 4
            MSS Value: 1460
        TCP Option - SACK permitted
            Kind: SACK Permitted (4)
            Length: 2
        TCP Option - Timestamps: TSval 1437226887, TSecr 0
            Kind: Time Stamp Option (8)
            Length: 10
            Timestamp value: 1437226887
            Timestamp echo reply: 0
        TCP Option - No-Operation (NOP)
            Kind: No-Operation (1)
        TCP Option - Window scale: 7 (multiply by 128)
            Kind: Window Scale (3)
            Length: 3
            Shift count: 7
            [Multiplier: 128]
    [Timestamps]
```

3.

A.

```
popi@DESKTOP-91RNSVG:/mnt/d/Facultate/an4/sem2/ep/labs/lab6/pwru-linux-amd64$ sudo !!
sudo ./pwru
[sudo] password for popi:
2025/04/11 08:45:00 Attaching kprobes (via kprobe)...
1806 / 1806 [----------------------------------------------------------] 100.00% 237 p/s
2025/04/11 08:45:07 Attached (ignored 57)
2025/04/11 08:45:07 Listening for events..
SKB                CPU PROCESS          NETNS      MARK/x        IFACE       PROTO  MTU
   LEN   TUPLE FUNC
0xffff8eb0161fd100 7   <empty>:8483     4026531840 0             eth0:2      0x0800 150
0  67    172.25.0.1:5353->224.0.0.251:5353(udp) ip_sabotage_in
0xffff8eafc88f5e00 6   <empty>:8483     4026531840 0             eth0:2      0x86dd 150
0  87    [fe80::563c:5c82:daec:6f1a]:5353->[ff02::fb]:5353(udp) ip_sabotage_in
0xffff8eafc88f5e00 6   <empty>:8483     4026531840 0             eth0:2      0x86dd 150
0  87    [fe80::563c:5c82:daec:6f1a]:5353->[ff02::fb]:5353(udp) ipv6_defrag
0xffff8eb0161fdb00 10  <empty>:8483     4026531840 0             eth0:2      0x0800 150
0  165   172.25.0.1:52175->239.255.255.250:1900(udp)           ip_sabotage_in
0xffff8eafc632de00 6   <empty>:8483     4026531840 0             eth0:2      0x86dd 150
0  125   [fe80::563c:5c82:daec:6f1a]:5353->[ff02::fb]:5353(udp) ip_sabotage_in
0xffff8eafc632de00 6   <empty>:8483     4026531840 0             eth0:2      0x86dd 150
0  125   [fe80::563c:5c82:daec:6f1a]:5353->[ff02::fb]:5353(udp) ipv6_defrag
0xffff8eb012198f00 7   <empty>:8483     4026531840 0             eth0:2      0x0800 150
0  105   172.25.0.1:5353->224.0.0.251:5353(udp)                 ip_sabotage_in
0xffff8eb012199900 7   <empty>:8483     4026531840 0             eth0:2      0x0800 150
```

```
8.8 example.com

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>> @8.8.8.8 example.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39504
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                   IN      A

;; ANSWER SECTION:
example.com.            190     IN      A       23.192.228.80
example.com.            190     IN      A       96.7.128.198
example.com.            190     IN      A       23.215.0.136
example.com.            190     IN      A       23.192.228.84
example.com.            190     IN      A       96.7.128.175
example.com.            190     IN      A       23.215.0.138

;; Query time: 29 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Apr 11 08:47:57 EEST 2025
;; MSG SIZE  rcvd: 136
```

```
[sudo] password for popi:
2025/04/11 08:46:57 Attaching kprobes (via kprobe)...
1806 / 1806 [----------------------------------------------------------] 100.00% 34 p/s
2025/04/11 08:47:50 Attached (ignored 57)
2025/04/11 08:47:50 Listening for events..
SKB              CPU PROCESS         TIMESTAMP        NETNS      MARK/x        IFACE      PROTO  MTU   LEN   TUPLE FUNC
0xffff8eb232dfb000 11  <empty>:8880     0                4026531840 0             0          0x0800 1500  80    172.25.2.72:36621->8.8.8.8:53(udp) nf_conntrack_in
0xffff8eb232dfb000 11  <empty>:8880     43053            4026531840 0             0          0x0800 1500  80    172.25.2.72:36621->8.8.8.8:53(udp) ip_output
0xffff8eb232dfb000 11  <empty>:8880     34256            4026531840 0             eth0:2     0x0800 1500  94    172.25.2.72:36621->8.8.8.8:53(udp) netvsc_select_queue
0xffff8eb232dfa100 11  <empty>:8691     0                4026531840 0             eth0:2     0x0800 1500  164   8.8.8.8:53->172.25.2.72:36621(udp) inet_gro_receive
0xffff8eb232dfa100 11  <empty>:8691     14528            4026531840 0             eth0:2     0x0800 1500  164   8.8.8.8:53->172.25.2.72:36621(udp) ip_rcv_core
0xffff8eb232dfa100 11  <empty>:8691     11973            4026531840 0             eth0:2     0x0800 1500  164   8.8.8.8:53->172.25.2.72:36621(udp) nf_conntrack_in
0xffff8eb232dfa100 11  <empty>:8691     3968             4026531840 0             eth0:2     0x0800 1500  164   8.8.8.8:53->172.25.2.72:36621(udp) nf_checksum
0xffff8eb232dfa100 11  <empty>:8691     11833            4026531840 0             eth0:2     0x0800 1500  164   8.8.8.8:53->172.25.2.72:36621(udp) fib_validate_source
0xffff8eb232dfa100 11  <empty>:8691     16462            4026531840 0             eth0:2     0x0800 65536 144   8.8.8.8:53->172.25.2.72:36621(udp) udp_rcv
0xffff8eb232dfa100 11  <empty>:8691     9168             4026531840 0             eth0:2     0x0800 1500  136   8.8.8.8:53->172.25.2.72:36621(udp) __udp_enqueue_schedule_skb
0xffff8eafc632d300 6   <empty>:8893     0                4026531840 0             0          0x0000 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) udpdl_hwcsum
0xffff8eafc632d300 6   <empty>:8893     4669             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) ip_send_skb
0xffff8eafc632d300 6   <empty>:8893     7845             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) __ip_local_out
0xffff8eafc632d300 6   <empty>:8893     7324             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) nf_hook_slow
0xffff8eafc632d300 6   <empty>:8893     4749             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) ipv4_conntrack_defrag
0xffff8eafc632d300 6   <empty>:8893     1503             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) ipv4_conntrack_local
0xffff8eafc632d300 6   <empty>:8893     4689             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) nf_conntrack_in
0xffff8eafc632d300 6   <empty>:8893     7545             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) nf_conntrack_udp_packet
0xffff8eafc632d300 6   <empty>:8893     5711             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) nf_route_table_hook4
0xffff8eafc632d300 6   <empty>:8893     7915             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) ttl_tg
0xffff8eafc632d300 6   <empty>:8893     3918             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) skb_ensure_writable
0xffff8eafc632d300 6   <empty>:8893     1202             4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) ttl_tg
0xffff8eafc632d300 6   <empty>:8893     541              4026531840 0             0          0x0800 1500  80    172.25.2.72:39480->8.8.8.8:53(udp) skb_ensure_writable
```

```
popi@DESKTOP-91RNSVG:/mnt/d/Facultate/an4/sem2/ep/labs/lab6/pwru-linux-amd64$ dig @8.8.8.8 example.com

; <<>> DiG 9.18.30-0ubuntu0.22.04.2-Ubuntu <<>> @8.8.8.8 example.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 39504
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 6, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;example.com.                   IN      A

;; ANSWER SECTION:
example.com.            190     IN      A       23.192.228.80
example.com.            190     IN      A       96.7.128.198
example.com.            190     IN      A       23.215.0.136
example.com.            190     IN      A       23.192.228.84
example.com.            190     IN      A       96.7.128.175
example.com.            190     IN      A       23.215.0.138

;; Query time: 29 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Fri Apr 11 08:47:57 EEST 2025
;; MSG SIZE  rcvd: 136
```

B.

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
137  1  DESKTOP-91RNSVG (172.25.0.1)  2.646 ms  2.330 ms  2.240 ms
140  2  192.168.85.152 (192.168.85.152)  33.002 ms  32.891 ms  32.835 ms
 3  XiaoQiang.lan (192.168.31.1)  32.919 ms  32.625 ms  32.749 ms
 4  192.168.0.1 (192.168.0.1)  32.514 ms  32.418 ms  32.314 ms
 5  10.0.2.58 (10.0.2.58)  33.526 ms  32.060 ms  31.981 ms
16411 1559.237006077 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 Who has 172.25.0.1? Tell 172.25.2.72
16412 1559.237276165 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 172.25.0.1 is at 00:15:5d:ec:55:f5
 6  10.243.116.97 (10.243.116.97)  30.462 ms  29.972 ms  29.832 ms
183  7  10.220.202.70 (10.220.202.70)  38.580 ms  33.152 ms  32.975 ms
156  8  10.220.148.208 (10.220.148.208)  32.827 ms 10.220.131.97 (10.220.131.97)  32.755 ms16413 1562.99662354
7 Microsof_ec:55:f5 → Microsof_ae:78:90 ARP 42 Who has 172.25.2.72? Tell 172.25.0.1
16414 1562.996679957 Microsof_ae:78:90 → Microsof_ec:55:f5 ARP 42 172.25.2.72 is at 00:15:5d:ae:78:90
 10.220.131.99 (10.220.131.99)  32.676 ms
 9  10.221.100.14 (10.221.100.14)  34.307 ms  34.243 ms  34.174 ms
10  72.14.216.212 (72.14.216.212)  34.083 ms 209.85.168.182 (209.85.168.182)  47.242 ms  47.131 ms
11  * * *
208 12  142.251.228.31 (142.251.228.31)  52.995 ms 209.85.255.243 (209.85.255.243)  52.949 ms dns.google (8.8.
8.8)  37.727 ms
218 @DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ ip route get 8.8.8.8
8.8.8.8 via 172.25.0.1 dev eth0 src 172.25.2.72 uid 1000
    cache
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ sudo tcpdump -i any -n -vv 'udp port 53 and host 8.8.8.8' -c 1
tcpdump: data link type LINUX_SLL2
tcpdump: listening on any, link-type LINUX_SLL2 (Linux cooked v2), snapshot length 262144 bytes
```

```
popi@DESKTOP-91RNSVG:/mnt/c/Users/Popi/Desktop$ uname -a
Linux DESKTOP-91RNSVG 5.15.133.1-microsoft-standard-WSL2 #1 SMP Thu Oct 5 21:02:42 UTC 2023 x86_64 x86_64 x86_
64 GNU/Linux
```

1. Pachetul pornește din WSL2 (172.25.2.72) și e direcționat către gateway-ul 172.25.0.1
2. Se face schimb de mesaje ARP pentru a afla adresa MAC a gateway-ului
3. Pachetul traversează routerele locale:
    - 192.168.85.152
    - 192.168.31.1 (XiaoQiang.lan)
    - 192.168.0.1
4. Intră în rețeaua furnizorului de internet prin 10.0.2.58 și traversează mai multe routere (10.243.x.x, 10.220.x.x)
5. Ajunge la infrastructura Google prin 72.14.216.212
6. În final, ajunge la serverul DNS Google (142.251.228.31 / 8.8.8.8) după aproximativ 38 ms

Tot timpul, pachetul UDP (port 53) rămâne încapsulat în IP, purtând interogarea DNS către serverul Google.

Feedback

